

Fairness-Aware Adaptive Routing for Quantum Networks

Responsible Data Science Analysis — Focus Area: Fairness — [WRITE: your analysis subtitle here — one line naming the specific angle, e.g., service-equity risks when entanglement routing is optimized for aggregate efficiency]

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Abstract

[WRITE (60–80 words, optional): one sentence on the capstone, one on the fairness concern, one on your mitigation stance, one on the venue-norm framing (limitations/broader-impact style).]

1 Context

[WRITE: Situate the artifact precisely. Cover, in order: (1) The system: a quantum-network controller that selects entanglement routes and allocates limited qubits under stochastic-to-adaptive threat regimes, evaluated by an inherited threat-aware bandit spine (the graduate-assistant quantum-routing work), now extended with fairness-aware mediation (your DSCI602 capstone). (2) The decision the system influences: which flows get successful, timely service when routes, qubits, and reliable context are scarce. (3) The “groups”: flows / service classes differentiated by demand, priority, and context quality — these are your equity units. (4) Stakeholders: tenants/applications sharing the network whose service depends on the controller’s allocation choices. (5) Why fairness is at stake: the evaluation objective is aggregate Oracle-normalized efficiency, and inherited evidence already shows non-obvious interactions (a validated replay slice where efficiency falls then recovers across capacity scales) — proof that aggregate metrics can hide regime-dependent behavior. (6) OPTIONAL specificity boost: quantum networks already require scheduling scarce capacity across simultaneous requests, and prior work frames this as a fair-sharing problem [Cicconetti et al. 2023; Li et al. 2021] — one sentence citing this precedent shows the fairness concern is native to quantum-network resource allocation, not imported from classical ML. Cite the venue-norm framing once here: NeurIPS checklist / ACL limitations practice [ACL Rolling Review 2023; Gibney 2020; NeurIPS 2023].]

2 Risk Analysis

2.1 Where disparity enters the pipeline

[WRITE: Map Suresh & Gutttag’s sources-of-harm lifecycle [Suresh and Gutttag 2021] and the Mehrabi survey’s bias taxonomy [Mehrabi et al. 2021] onto YOUR pipeline concretely: measurement (which flow outcomes get recorded — success, latency, reward today; demand, allocated qubits, retries/starvation only planned), evaluation (aggregate efficiency as the headline metric), and deployment (threat regimes that shift the operating point). Name the specific stage, not just the category. Gap to close: a bandit controller also has a sequential feedback risk that the lifecycle/taxonomy papers

alone do not cover — underexplored routes or cohorts generate less reward evidence, which can further lower their future selection. Cite [Joseph et al. 2016] (fairness changes the regret cost of learning under uncertainty) and [Chen et al. 2020] (contextual allocation with minimum service rates) to name this feedback-loop mechanism explicitly.]

2.2 Which fairness definition fits — and why none is free

[WRITE: Present demographic-parity-style per-class service gaps versus error-rate parity in your setting [Barocas et al. 2023; Hardt et al. 2016]; then use the impossibility results [Chouldechova 2017; Kleinberg et al. 2017] to justify your capstone’s actual design stance: measure a utility–disparity frontier across a $\lambda, \gamma, \epsilon$ mediation sweep rather than assert one metric. This is your deepest technical move — the frontier exists precisely because the definitions cannot all hold at once. OPTIONAL formal anchor: state a concrete disparity/access metric instead of leaving the gap abstract, e.g. demand-normalized access $A_g(T) = \sum_t Q_{g,t} / \sum_t D_{g,t}$ and its max-gap Δ_A , or Jain’s fairness index over group-normalized access [Jain et al. 1984]. A named formula is a concrete, citable specificity gain over describing the frontier only in words.]

2.3 Concrete failure modes and who bears them

[WRITE: 2–3 named failure modes, each with the affected group: (a) aggregate gains concentrating in high-demand classes while low-demand or poor-context flows absorb retries/starvation under escalating (structured/adaptive) threats; (b) “fairness through unawareness” — the current post-run records lack the fields needed to even see disparity (cite the caution via [Barocas et al. 2023]); (c) deployment drift — a policy tuned in one threat regime transferring poorly, echoing how audited real-world systems failed groups the aggregate metric hid [Buolamwini and Gebu 2018; Obermeyer et al. 2019]. Keep the quantum claims prospective; the two cited harms papers are your evidence that aggregate-optimized systems conceal group harm in general.]

3 Mitigations

[WRITE: Present mitigations as measurement-first, then intervention, then process — each tied to your actual architecture: (1) Measurement: per-class success/latency/resource-access/retry/starvation gaps computed from enriched outcome records (your D1 schema work) — cite measurement/auditing practice [Barocas et al. 2023;

Mehrabi et al. 2021]. (2) Intervention: the planned mediation score $S_t^{fair}(a) = U_t(a) - \lambda \max(0, \hat{\Delta}_t(a) - \epsilon) - \gamma C_t(a)$ attached at a decided seam, evaluated off-vs-on (your D2) — frame as a fairness–utility trade-off dial, cite [Barocas et al. 2023; Hardt et al. 2016]. Note the mediator keeps the base policy separable and auditable rather than replacing it, matching the design pattern in [Chen et al. 2020]. (3) Process: datasheets/model-card-style documentation for experiment artifacts and the contribution ledger separating inherited from new work (your D5) [Gebru et al. 2021; Mitchell et al. 2019]. (4) Residual risk: impossibility means some disparity measure always survives [Chouldechova 2017; Kleinberg et al. 2017]; mediation can cost utility in some regimes — that is why the frontier is measured, not assumed.]

4 Limitations

[WRITE, in the ACL-limitations voice [ACL Rolling Review 2023]: (a) the analysis is prospective: live mediation and fairness-mediated experiments are planned, so no fairness result is claimed yet; (b) evidence comes from a simulator that does not yet model hardware-anchored timing/resource contention; (c) service classes are a design choice — poorly chosen classes can encode the wrong groups; (d) two pages cannot treat security/privacy intersections (poisoning of threat feedback, privacy of flow telemetry) except by pointer.]

References

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